From the statistical point of view, a more modest fluctuation that produced some order but not as much as we currently see would be *far* more likely. Moreover, since on average entropy is on the rise, Boltzmann's reasoning suggests that it would be much more likely that everything we see today *just now* arose as a rare statistical jump to lower entropy. Recall the reason: the farther back the fluctuation happened, the lower the entropy it would have had to attain (entropy starts to rise after any dip to low entropy, as in Figure 6.4, so if the fluctuation happened yesterday, it must have dipped down to yesterday's lower entropy, and if it happened a billion years ago, it must have dipped down to that era's even lower entropy). Hence, the farther back in time, the more drastic and improbable the required fluctuation. Thus, it is much more likely that the jump just happened. But if we accept this conclusion, we can't trust memories, records, or even the laws of physics that underlie the discussion itself—a completely intolerable position.

The tremendous advantage of the inflationary incarnation of Boltzmann's idea is that a *small* fluctuation early on—a *modest* jump to the favorable conditions, within a *tiny* nugget of space—inevitably yields the huge and ordered universe we are aware of. Once inflationary expansion set in, the little nugget was *inexorably* stretched to scales at least as large as the universe we currently see, and hence there is no mystery as to why the universe didn't cut corners; there is no mystery why the universe is vast and is populated by a huge number of galaxies. From the get-go, inflation gave the universe an amazing deal. A jump to lower entropy within a tiny nugget of space was leveraged by inflationary expansion into the vast reaches of the cosmos. And, of utmost importance, the inflationary stretching didn't just yield any old large universe. It yielded *our* large universe—inflation explains the shape of space, it explains the large-scale uniformity, and it even explains the "smaller"-scale inhomogeneities such as galaxies and temperature variations in the background radiation. Inflation packages a wealth of explanatory and predictive power in a single fluctuation to low entropy.

And so Boltzmann may well have been right. Everything we see may have resulted from a chance fluctuation out of a highly disordered state of primeval chaos. In this realization of his ideas, though, we can trust our records and we can trust our memories: the fluctuation did not happen just now. The past really happened. Our records are records of things that